Location-Based Service Insights: A Comprehensive Review of Data Mining Objects

Bayu Yanuargi
Department of Information Technology
Universitas Amikom Yogyakarta
Sleman, Indonesia
bayyanmc@gmail.com

Ema Utami
Department of Information Technology
Universitas Amikom Yogyakarta
Sleman, Indonesia
ema.u@amikom.ac.id

Kusrini
Department of Information Technology
Universitas Amikom Yogyakarta
Sleman, Indonesia
kusrini@amikom.ac.id

Arli Aditya Parikesit,
Bio Informatika
Indonesia International Institute for Life Sciences
Jakarta, Indonesia
arli.parikesit@i3l.ac.id

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***Abstract*— Location-based services are increasingly popular across diverse industries, spanning transportation, logistics, and tourism. Extensive studies have delved into this field, exploring facets such as application, performance enhancement, and security vulnerabilities that may endanger users. This systematic literature review, encompassing the years 2020 to 2024 and sourced from IEEE Xplore, investigates the prevalent objects and data types in location-based services. The paper's methodology for selecting articles extends beyond search keywords, incorporating advanced terms from abstracts. This dual criteria led to the identification of fifty papers based on abstract keywords. The systematic review categorized papers based on objects and data, revealing seven research categories: privacy protection data, urban planning, POI data, mobility, farming, tourism, and indoor maps. Data types were classified into two groups: social media data and non-social media data. The analysis reveals that effective utilization of location-based data is pivotal in enhancing various facets of daily life, streamlining processes, and contributing to more informed decision-making. However, the study also highlights the imperative need to address data privacy concerns vigilantly. By shedding light on these aspects, this paper aims to contribute to the ongoing discourse in the field of LBS, emphasizing the balance between technological advancement and ethical data handling.**

Keywords—location-based service, location social media, location recommendation, location prediction, location data

I. Introduction

The cellular era that we are currently facing has changed the science of information technology, especially how the location tracking system has been developed rapidly including GPS tracking for the people or objects moving [1]. Human activities tracked by the location tracking system can produce highly valuable location-based data. The increase in location-based services significantly attracted the sciences and researchers to conduct several research related to location-based services (LBS) in any discipline [2].

The success of an LBS is closely related to its ability to effectively aggregate and analyze location data. Due to the daily activities of millions or even billions of users, LBS can produce big data that always increases in terms of velocity, volume, value, variety, and veracity. In another sector of usage based on human location activities, the location big data can describe the structure and dynamics of cities as well as human mobility or movement, as well as provide empirically validated solutions to various urban challenges, such as air pollution and traffic congestion. Therefore, LBS data is now recognized as a key component in sustainable and smart city development [2].

LBS uses a smart device sensor, usually using three GPS features, a Phone Signal, and the nearest telecommunications towers to make it able to provide the position or location estimations of the users, outside or inside the building. Aligned with the awareness of the Internet of things by the technology users impacted the daily activities depending on how effective and efficient the provider of the services and supporting system to gain the near perfect experiences. In the growing markets that keep growing and the demand for real-time accuracies of the location or position estimation, industries racing on the competition in technological research and development to improve their services or to build new features for their products to get better performance on the LBS sector [3].

One technology that uses LBS data directly and is used by users in daily activities is digital map technology such as Google Maps. By using LBS technology Google Maps allows people to navigate and find the shortest and most comfortable route to their desired destination. According to a recent survey, Google Maps has gained almost 64 million users. Additionally, it has included new features like street view, hospital locations, cafes, police stations, and many more useful features. The algorithms, techniques, and technology used by Google Maps are state-of-the-art and highly advanced [4].

Several applications of Location-Based Services (LBS) data in digital maps are widely employed across various industries. One notable example is its integration within the logistics and transportation sector, aimed at addressing the challenges associated with optimizing the logistical industry structure amid the rapid growth of e-commerce in today's interconnected logistics networks. Additionally, the utilization extends to the mechanism of establishing locations by harnessing Point of Interest (POI) data through the Gaode Map API platform. This demonstrates the multifaceted role that LBS data plays in enhancing efficiency and adaptability within different sectors, exemplified by its strategic application in tackling the

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evolving landscape of the logistics industry amid the dynamic developments in e-commerce [5].

Navigation services that used LBS to ensure it can be used properly have been popular applications recently and can be installed on mobile devices to provide the routing information, point of interest information, and current location of the users. With huge usage, LBS keeps facing challenges including the communication process between users and the service providers using mobile phones to make quick reactions on any decision making effectively in a certain time range. Those situations showed several basic challenges for this technology globally which will need to be monitored and evaluated frequently, so the economic impact of the LBS can be increased and user experience can be managed [6].

In the context of Location-Based Services (LBS), the primary objective of this paper is to undertake a comprehensive collection and analysis of papers or studies related to LBS data. This involves detailing the characteristics of LBS data, methodologies employed, the issues or objectives addressed in each paper, and highlighting the significant role of LBS data presented in each paper encountered within the timeframe spanning from 2020 to 2024. A profound understanding of LBS data not only proves beneficial in optimizing existing services but also opens doors to the development of innovations that can enhance the quality of life and efficiency across various aspects of life.

This research is expected will make a significant contribution to the literature on LBS data, providing readers with insights into the potential and challenges associated with location data management. Furthermore, it aims to stimulate further thinking in understanding how LBS data can be optimally utilized to support the advancement of information technology and various related sectors. Thus, through the exploration of LBS data, this paper endeavors to establish a solid foundation for further development in this field.

II. METHODS

This paper's research method is about systematic literature review (SLR), namely a systematic and structured research method for identifying, interpreting, and comparing all relevant research that has been conducted in a particular field of knowledge. There are many reasons to conduct a SLR. The most common reasons include: summarizing existing evidence about a treatment or technology, for example summarizing empirical evidence about the benefits and limitations of a particular agile method; identifying gaps in current research to suggest areas that need further research; and providing a framework or background against which to situate new research activities appropriately [7]. Based on Kitchenham B, this research begins with the following research questions:

- RQ1: What LBS objects were studied in each paper?
- RQ2: What types of location-based data were used in LBS-related research?

The purpose of the above research question is to understand how is LBS technology growing within 2020 – 2024 by identifying the paper object. The trend of the used dataset will also be interesting to review, since a huge increase of connected locations data usage in social media such as Foursquare, Twitter, Gowalla, Brightkite, Instagram, etc will be impacting the direction of the LBS technology.

In this paper, the author collected research papers related to LBS in the time range 2020 – 2024, where the paper collection process was carried out at ieeexplore to obtain papers from research results that were quality and indexed by Scopus. The steps or processes carried out in the SLR can be seen in Figure 1 below.

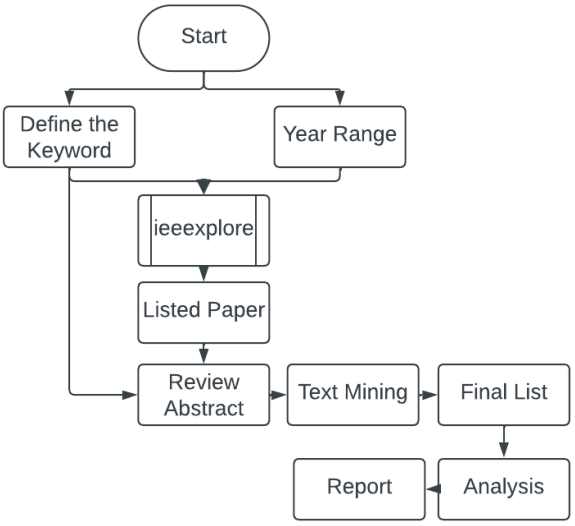


Fig 1. SLR Process Flow

A. Inclusion and Exclusion Criteria

In general, the steps for this literature review are after obtaining several papers based on keywords, the paper search process is carried out on the website https://ieeexplore.ieee.org/. The results of a search on the website found around 1400 open-access papers. The inclusion criteria in this SLR can be seen in the points below, while exclusion criteria are other than those mentioned in the following point:

- Search process using keywords location-based services, location-based data mining, and POI.
- The time range used is 2020 – 2024 and only for journal-type papers
- The third inclusion criterion is this paper only uses the open-access paper

The second phase of the filtering was conducted after the first filtering based on the three inclusion criteria above, where the top 350 open-access papers were used that were relevant to the given keywords and within the time range. The distribution of the papers based on the published year has the highest number of LBS papers in 2021 and fewer papers will be published in 2023 and 2024, the lowest number in 2024 because this SLR wrote in the beginning of 2024.

The second phase of filtering is conducted Based on the existing abstract from the selected paper above, a

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text-mining process is carried out to find papers with abstracts that mention the keywords used. This is done to further focus SLR on papers that focus on location-based services. Based on the text-mining process using keywords location-based, location-based, lbs, lbsn, place, places. Used of filtering based on abstract aiming to get more focusing paper on LBS. The filtering process related to the collected papers that involved two phases of inclusion criteria can be seen in below flow chart Fig 2 below.

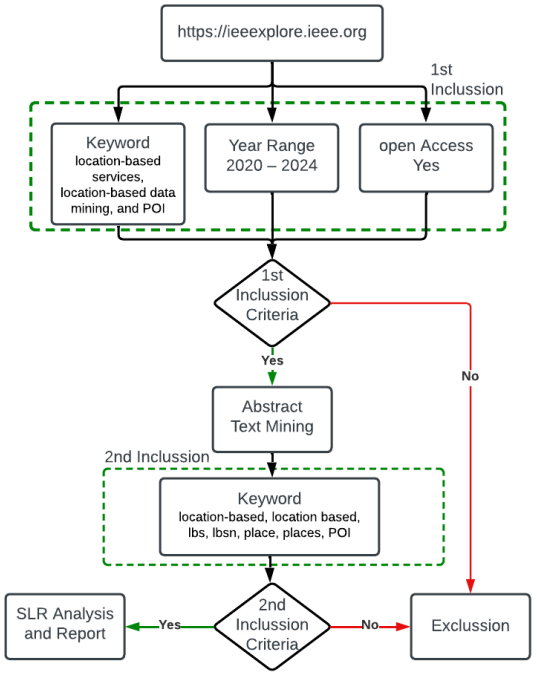


Fig 2. Inclusion and Exclusion Process on Paper Selection

III. RESULT AND DISCUSSION

Based on the results of simple statistical analysis, at least seven research objects were obtained from the 50. The nine research

A. Research Object or Topic on LBS

Based on the results of simple statistical analysis, at least seven research objects were obtained from the 50. The nine research objects are related to increasing the security of privacy data for users of LBS technology, analyzing user mobility data, implementing LBS data in urban planning, farming, and tourism, increasing accuracy in indoor maps products and the last one is the acquisition or prediction of POI data. In Figure 3 below we can see the composition of research objects from the 50 papers reviewed with the largest paper composition of the use of LBS data in the acquisition and prediction of POI data.

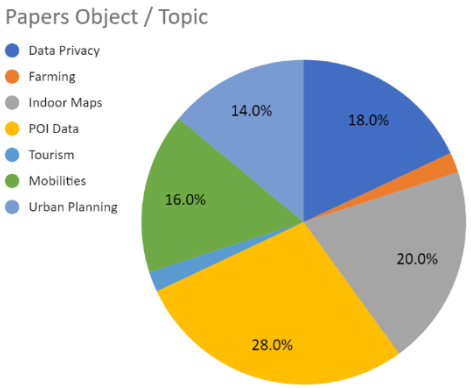


Fig 3. Papers Focus Object

By using descriptive analysis based on the graph above, it can be seen that during 2020-2024 research related to LBS is quite diverse and each paper has quite interesting discussion objects that can contribute to further LBS research, especially as LBS technology continues to develop and is used by many industries and provide significant added value to related industrial services. In table II below, it is explained in more detail regarding each object discussed in the 50 papers analyzed.

TABLE I. STUDY DETAIL ON EACH OBECJT

Object	Details
Data Privacy	Based on the review process, eight papers focus on protecting data privacy, of which four involve users to improve security, while the other four utilize technology without directly involving users [8][9][10][11][12][13][14][15]. Apart from the data, only one paper focuses on the process of detecting false locations which can affect data security [16].
Farming	Use the location-based data on a very detailed scope to detect the pest location [17].
Indoor Maps	Most of the LBS papers related to indoor mapping state the challenges of GPS / positioning accuracy in the building environment, so it has a big impact on LBS accuracy, for this reason, six papers out of ten discuss issues related to increasing LBS accuracy in indoor environments [18][19][20][21][22][23]. One paper discusses how to complete tracking data in buildings that is incomplete or missing due to weak GPS signals in buildings [24]. Three other papers conducted research related to the process of optimizing LBS data collection in buildings, although the final results still discussed the completeness and accuracy of the data collected [25][26][27].
Mobilities	Papers with mobility objects are quite diverse, starting from improving LBS accuracy in collecting mobility data including trajectories ranging from user involvement in determining LBS networks to models for recommending the best LBS networks [28][29]. Three papers conducted research related to polar movement or human mobility to predict and recommend vehicle mobility routes that are expected to provide the best route for cellular users [30][31][32] Two papers examine user mobility objects and then categorize and classify user mobility or trajectory data [33][34]. Meanwhile, the last paper carried out a labeling process to provide anonymous labels to mobility data before it was used for further research [35].
Tourism	The paper in this research object processes check-in data carried out by location-based social network (LBSN) users and predicts the mobility patterns of tourists visiting new countries with or without records of previous visits [36].
Urban Planning	In this research object, three papers discuss predictions and recommendations for public facilities in urban areas, starting from traffic control, parking lot predictions, next-location predictions, location-based advertising, and recommendations for interesting places [37][38][39]. In urban planning, several papers discuss the classification of urban area functions and

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	also the differentiation of urban and rural areas by approaching urban density predictions [40][41][42][43].
POI Data	Papers with research objectives related to POI mostly discuss recommendations for users in determining the location of their next visit, by looking at visit patterns and user mobility [44][45][46][47][48][49][50][51]. Giving recommendations is almost the same as prediction, but only two papers directly mention the prediction approach, whether in terms of subsequent visits or predictions to provide recommendations for a new POI [52][53][54]. The other two papers focus more on increasing point accuracy based on interactions on social media and also accuracy in feature recognition in different texts[55][56] One of the studies discusses an effective POI data extraction mechanism from several sources such as Google Maps and social media [57].

B. Types of Location-Based Data

Recently, social media has dominated in providing LBS data, in this SLR the author tried to group papers based on the type and source of data used in the 50 papers reviewed. The process of reviewing the datasets in each paper is quite simple, namely by identifying what LBS datasets are used and the sources of obtaining these datasets. This will be very useful for further research as a reference for collecting appropriate LBS data. The focus of the data grouping carried out is on types of social media and non-social media data, to explore the potential of social media in providing LBS data that is not limited by space and time. This will of course be different from non-social media data, which is of course limited and more difficult to obtain. The composition of the Social Media and Non-Social Media datasets can be seen in Figure 4 below, where the number of papers with data from social media is 19 papers and the number of papers with data from direct collection in the field (non-social media) is 31 papers.

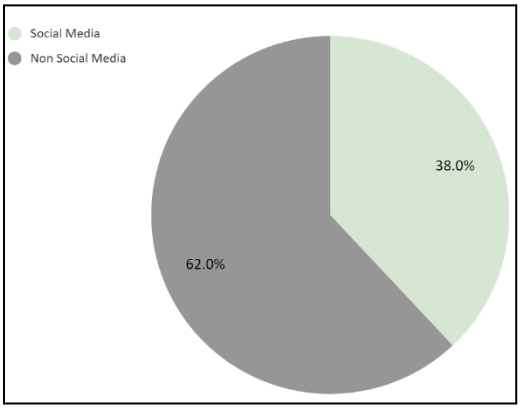


Fig 4. Papers Data Types

Through a simple analysis of the dataset used, a significant relationship pattern was found between the research object of each paper and the type of data used. The results of this analysis are documented in Table III, which shows the distribution of data types and their relationship to each data object. Understanding this relationship pattern provides deeper insight into the relevance between research objects and the characteristics of the related data types.

TABLE II. DATA TYPE ON EACH PAPER OBJECT

Object	Paper Number	
	Non-Social Media	Social media
Data Privacy	8	1
Farming	1	0
Indoor Maps	10	0
Mobilities	4	4
POI Data	3	11
Tourism	0	1
Urban Planning	5	2

Based on Table III above, we can see that papers that focus on indoor maps, farming, and data privacy use little or no social media data in their research. In the paper on object data privacy, there is more discussion regarding data security methods using data collected directly from GPS application use and direct data collection in the field. A detailed explanation of data use can be seen in Table IV below.

TABLE III. DATA TYPE ON EACH PAPER OBJECT

Object	Data Analysis
Data Privacy	One paper conducted an experiment using high-tech digital maps to capture longitude and latitude data to build anonymous sets and measure the impact of different privacy measures on service quality [8] [9]. Meanwhile, three papers directly use real users by tracking the movements of selected users and then involving users in determining the security level of an LBS-based application [10][11][12]. On the topic of privacy data, four papers use data obtained from second parties, namely in the form of archived data from user activities in the past. This will be useful for studying user habits and activity history so that security gaps can be found [13]. Only one paper in the field of data privacy involves social media data to obtain diverse data, where according to the paper this diversity of data has a high risk in data privacy security [14] [15]. One study used direct data from the field by taking direct measurements of the RSSI so that it could determine accuracy and find false locations[16].
Indoor Maps	All papers discussing indoor mapping agree that indoor maps have their challenges in terms of accuracy because in indoor environments GPS or cellphone signals will be disturbed by the presence of obstructing walls, so the use of GPS or cellphone signals will interfere with the accuracy of determining location inside the building. All papers use data taken directly in the field, whether using independent measurements or involving users, where the data collected is how good the WiFI signal reception (RSSI) is at each different location in a building. By using the RSSI data, researchers will be able to determine certain models or algorithms to increase the accuracy of determining location points [18][19][20][21][22][23] [24] [25][26][27].
Mobilities	In research related to human movement, one paper uses archival data obtained from a Microsoft project. The use of this archival data is expected to provide human movement patterns so that future predictions can be made. Four papers have successfully implemented data mining from social media to obtain human mobility data, where the use of social media data is expected to continue to provide unlimited and renewable data because the use of social media continues to grow, so the models built will continue to be developed based on continuously developing data [28] [29] [30] [32] [33] [34] [35]
POI Data	Most of the aims of papers with POI data research objects are recommendations and predictions, so most of the data used is social media data, where the use of social media data is obtained from user activity on several traveling applications

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	such as Foursquare, Yelp, etc. The use of social media data in POI recommendations and predictions is very important because apart from getting LBS data you will also get data in the form of reviews from social media users so that it can be used as a parameter in research to provide more accurate recommendations or predictions [44] [45] [46] [47] [48] [49] [50] [51] [54] [55] [57]. Apart from social media, two papers use archival data because they do not require direct review from users, the use of archival data or old data is more about discussing methods for correcting data if it is transformed into another form (Translation) and also developing methods in business intelligence, so that prioritizing use of existing data over data enrichment [52] [53] [56]
Urban Planning	In research related to urban planning, the use of LBS data is very important for spatial information, some papers use archival data and digital maps to present information and predictions related to city/regional planning [37][38] [39][40][43]. Meanwhile, two papers use social media data to zone and classify urban and rural areas using a spatial and temporal pattern mining approach [41] [42]
Farming	In agriculture, the LBS data used is obtained directly from the field using sensors installed in agricultural fields to map agricultural pests. The use of direct field data is due to the small scale of agriculture so the use of data measured directly in the field will be more accurate and better than data secondary [17]
Tourism	Predicting tourism mobility will be greatly influenced by the number of visits to tourist destinations and also influenced by visitor satisfaction, the use of social media data will be able to help make predictions more accurate, this is because tourists these days always record their activities and reviews of a tourist location. [36]

IV. CONCLUSION

In this systematic literature review paper, the analysis process focuses on two research questions. The first involves mapping the research object of each paper, while the second explores the type of dataset utilized by each paper to achieve its objectives.

Based on research results, in the 2020 - 2024 decade, there are seven research objects related to LBS, namely data privacy protection, indoor mapping, POI data, urban planning, farming, and tourism. The largest composition of research objects is found in papers that analyze POI data. Regarding the second research question, the type of data used is dominated by non-social media data, including for research objects that would be more optimal if using social media data but would prefer to use non-social media data.

The use of social media data like Foursquare, Twitter, Gowalla, Brightkite, Instagram, etc has the advantage of abundant and renewable data, but social media data has the disadvantage of being unstructured and requiring more management to be used. Meanwhile, non-social media data may have large amounts but in terms of time it comes from data that is older than social media, but the management process will be easier because most of it is structured data. Based on the results of the review, it is hoped that further research will be carried out to complement existing objects or add other more interesting objects by focusing on the application of social media data in real time.

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